Project Group A

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https://www.kaggle.com/code/ajacobo/more-macrocircuits

Macrocircuits

Energy Efficiency in RL control

Modified reward structures in MLP and RL

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Using wisely decided energy resources is very important in any system and biological systems have shown to achieve good performance at minimum energy cost.

To explore the implications of this in the control of the C. Elegans bio-system, we propose to modify the predefined reward function in order to cater for the minimization of the control effort.

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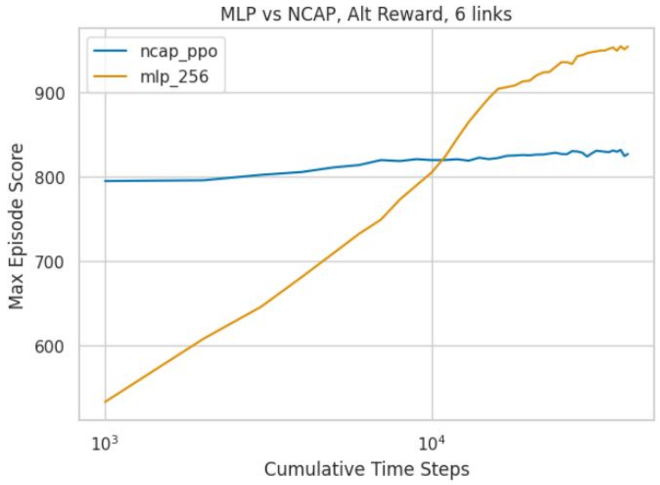
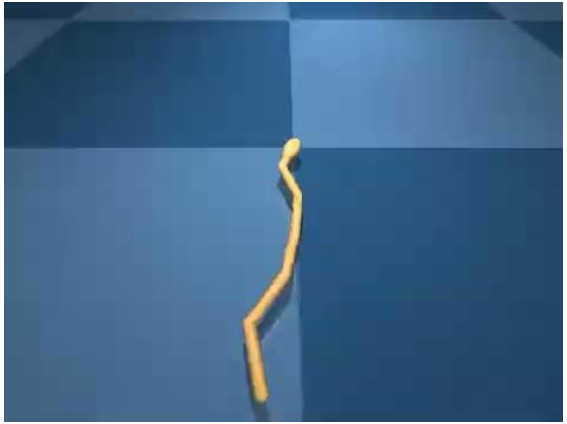
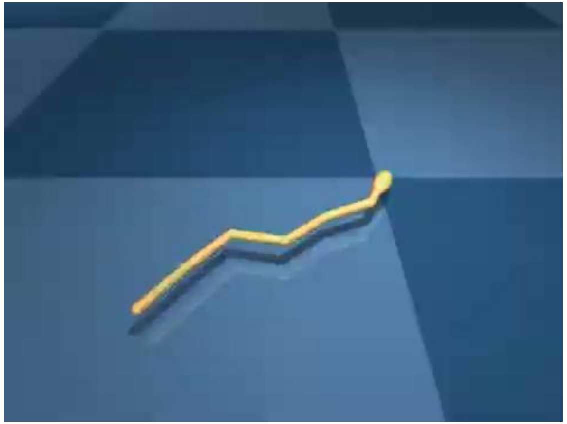
Testing the model performance for two alternative networks (**MLP and NCAP**) and two environment conditions. Notice that even though MLP seems to achieve better performance, NCAP already has “innate knowledge”.







Hypothesis

We're interested in studying the different models' motor behavior as well as performance and training efficiency when novel reward functions based on a different environment variable are introduced.

Question 1: What kind of performance and motor behavior will a MLP produce when it is trained to propel an agent toward a laterally placed target?

Question 2: Will NCAP maintain its performance of oscillating locomotion when trained to propel an agent toward a laterally placed target?

Question 3: How will the models perform when trained to minimize joint force exertion as well as propelling the agent forward?

We are focused on **comparing and contrasting the performance of the NCAP model with that of the MLP model** based on a newly defined reward function. The next steps heavily depend on the proper execution of the implementations of the NCAP and MLP models performance comparison based on the new reward function.

We modify the reward function, and observe the motor movements and oscillations of the artificial C. elegans organism. Analysis of the organism's joint movements and oscillatory characteristics allows to draw interesting conclusions.

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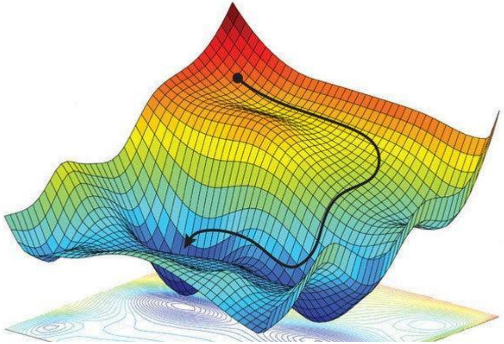
Reward Function Modification



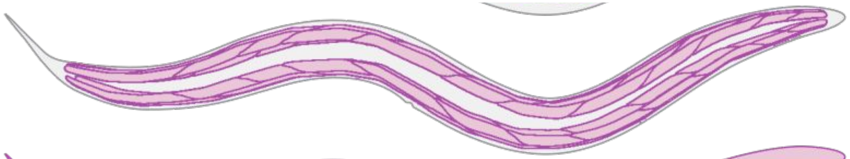
The new proposed reward function for the MLP case is multi-objective, where the control effort term $\sum \ddot{q}_i^2$ ded as follows:

$$M_r = \frac{[forward_{velocity} + \rho(1 - \ddot{q}'\ddot{q}/N)]}{2}$$

Where N = number of joints -1

where the modified reward $\|M_r\| \leq 1$ hence we reward highly low magnitude control actions.

It must be highlighted that the optimization problem becomes multi-objective, with opposing objectives, i.e. moving forward vs. using no energy, hence weights (in this case use $\rho \leq 1$) must be added to offset both objectives.



Similar to the original example, the new reward function will award partial points to the agent for every time step it moves forward. A second condition will award partial points for every time step the mean actuator applied force remains below 0.6 of maximum activation in either direction.

The choice of 0.6 is somewhat arbitrary, but it is based on C. elegans histology: every muscle cell is composed by 10 contractile units –sarcomeres– that turn that segment by approximately 6° each.

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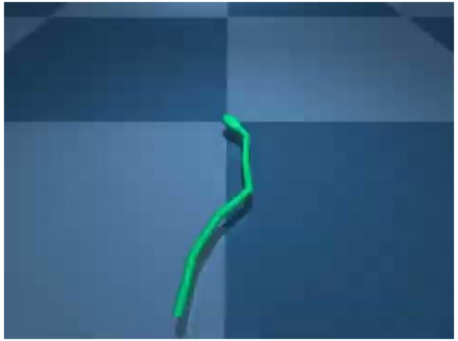
Reward Function Modification

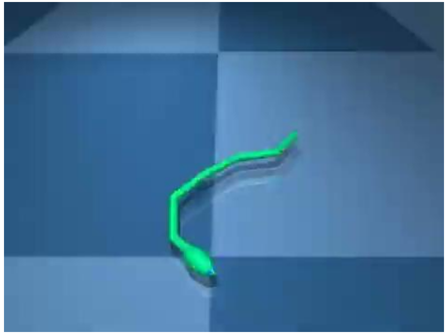
The MLP model was trained varying training time $step = \{1e^5, 2e^5, 3e^5\}$ and the weight $\rho = \{0, 0.5, 0.1\}$

The first conclusion is that the number of training steps has a strong influence on the performance; this becomes more noticeable for the modified task with ``high'' values of ρ

It was observed that a small training time (step = 1e5) and ρ = 0.5 produces an MLP control strategy that renders limited joint movement (large reward) but reduced forward velocity (low reward); <u>although our main objective is not reached, the result still gives a large reward.</u>



Step = 1e5, $\rho = 1$



Step = 1e5, $\rho = 0.1$

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Control Effort

After some simulation analysis, the step time and parameter ρ here set to 3e5 and 0.1 respectively.

We proceeded to experiment with the number of joints, producing controllers for 6, 12 and 24 joints for both M_r and the nominal reward (i.e. forward velocity).

We show the FFT and RMS value of motor actions in order to compare energy expenditure.

- In Table 1, it can be observed that for fewer links, M_r has lower RMS values; as the number of links increases, the RMS values get closer together.

	RMS value of motor action		
	Modified Reward	Pretrained	Delta (Pt - Mr)
M1	0,57737	0,68986	0,11249
M2	0,70235	0,92476	0,22241
M3	0,61534	0,78748	0,17214
M4	0,61677	0,93316	0,31640
M5	0,57786	0,92256	0,34470
	Modified Reward	Nominal Reward	Delta (Pt - Mr)
M1	0,74470	0,78717	0,04248
M2	0,79625	0,79989	0,00364
M3	0,74714	0,80533	0,05819
M4	0,74862	0,81077	0,06215
M5	0,71768	0,81387	0,09619
M6	0,74366	0,80364	0,05997
M7	0,75820	0,80086	0,04266
M8	0,70636	0,78892	0,08256
M9	0,73530	0,80463	0,06934
M10	0,65470	0,81110	0,15640
M11	0,64803	0,78840	0,14038
	Modified Reward	Nominal Reward	Delta (Pt - Mr)
M1	0,74256	0,75563	0,01306
M2	0,74652	0,79657	0,05005
M3	0,66389	0,76289	0,09900
M4	0,61833	0,72550	0,10717
M5	0,64940	0,73773	0,08833
M6	0,64562	0,76167	0,11605
M7	0,69981	0,78013	0,08032
M8	0,69454	0,78117	0,08664
M9	0,66780	0,80612	0,13832
M10	0,67400	0,80342	0,12942
M11	0,67226	0,74937	0,07712
M12	0,63464	0,78849	0,15384
M13	0,65126	0,79549	0,14424
M14	0,62813	0,70274	0,07461
M15	0,65782	0,62045	-0,03736
M16	0,72410	0,71387	-0,01024
M17	0,72534	0,78816	0,06282
M18	0,73942	0,78086	0,04144
M19	0,77388	0,77633	0,00244
M20	0,76136	0,78635	0,02499
M21	0,76199	0,77111	0,00912
M22	0,74081	0,74942	0,00860
M23	0,69188	0,76265	0,07077

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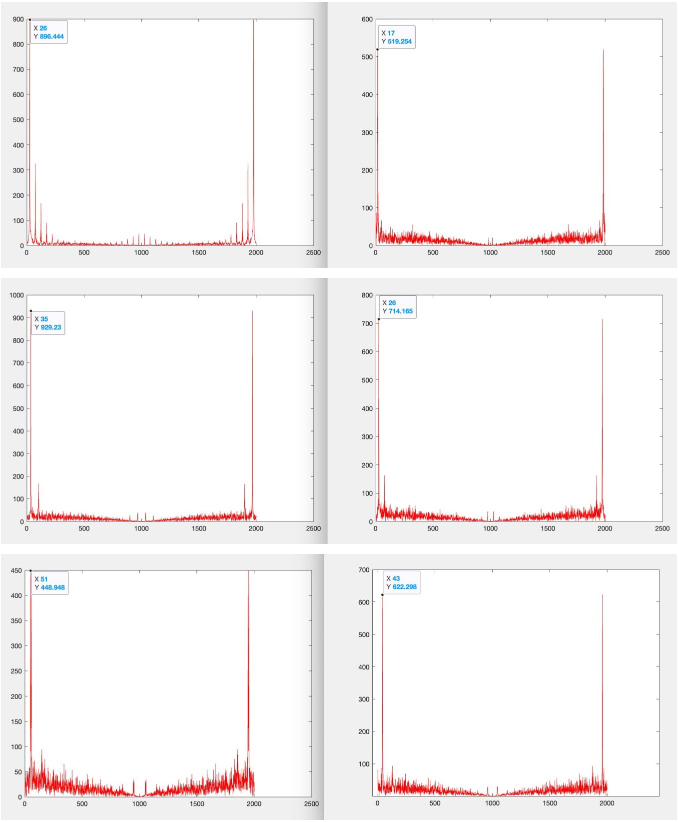
FFT analysis

After some simulation analysis, the step time and parameter ρ where set to 3e5 and 0.1 respectively.

We proceeded to experiment with the number of joints, producing controllers for 6, 12 and 24 joints for both M_r and the nominal reward (i.e. forward velocity).

We show the FFT and RMS value of motor actions in order to compare energy expenditure.

- Similarly, the magnitude of the first fourier moment (i.e. resonant frequency) is higher for the nominal reward function, but this gap seem to reduce as the number of joint is increased.



6 Joints

12 Joints

24 Joints

Nominal Reward

Modified Reward

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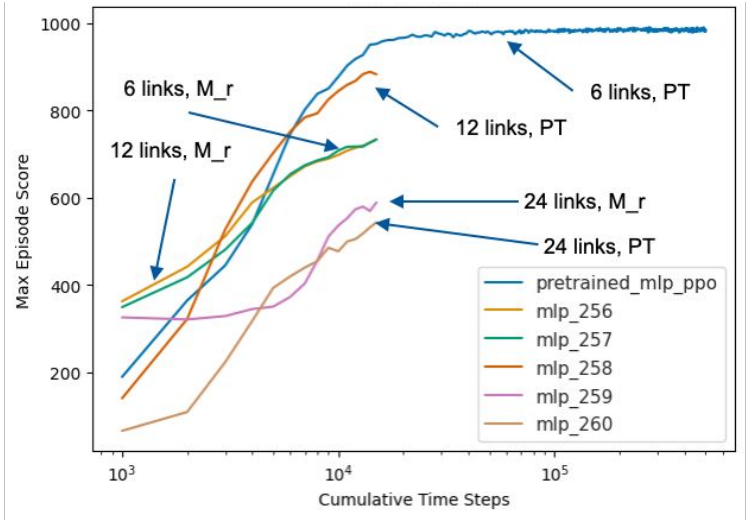
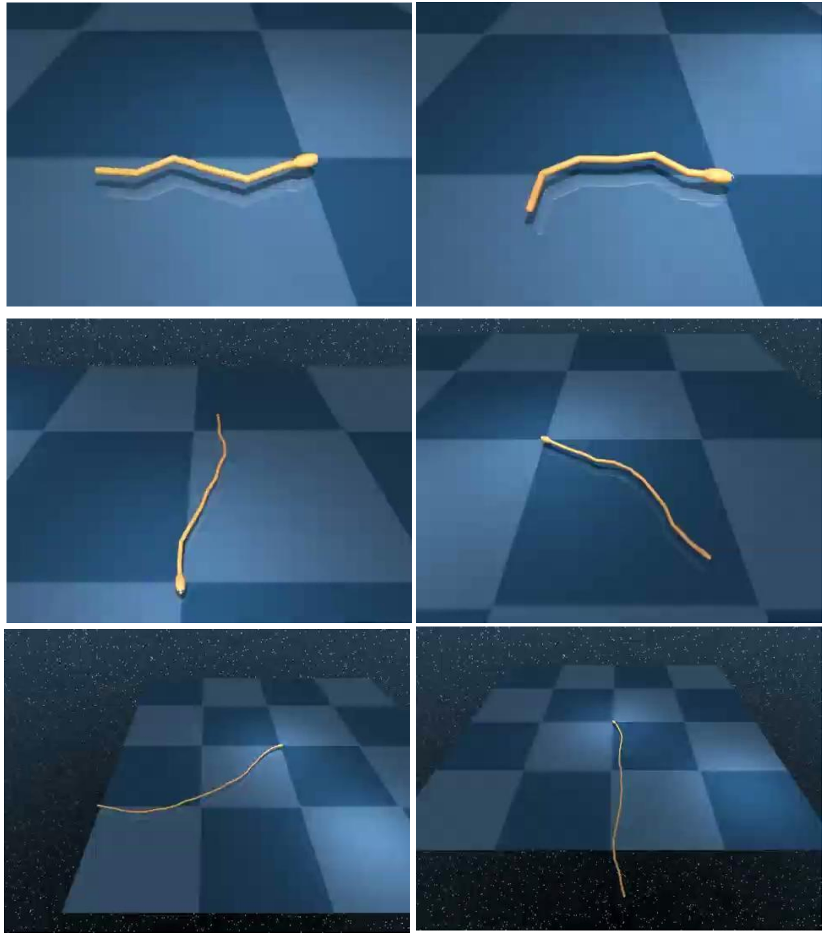
Performance Analysis (MLP)





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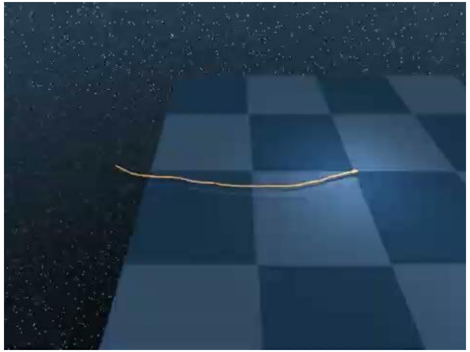
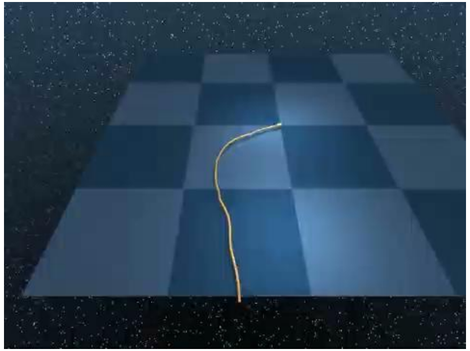
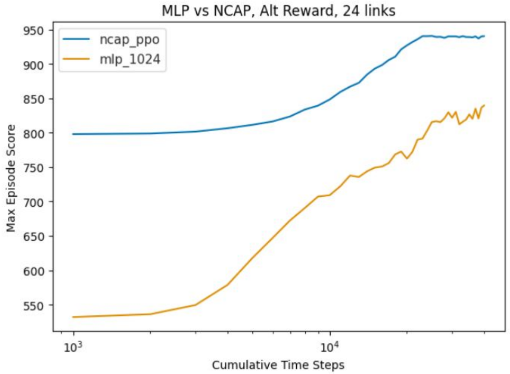
Performance analysis

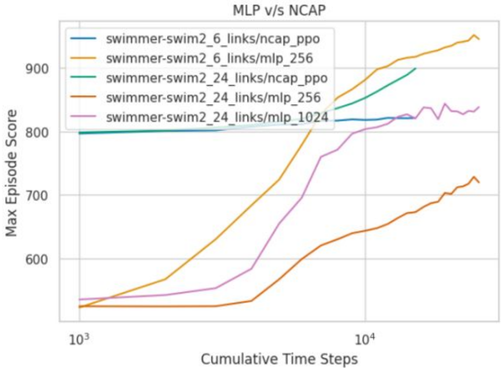




Fascinatingly, with the new reward function, a **bigger worm performed much better using NCAP**.



Wider MLP indeed does better; **but not as well as NCAP**



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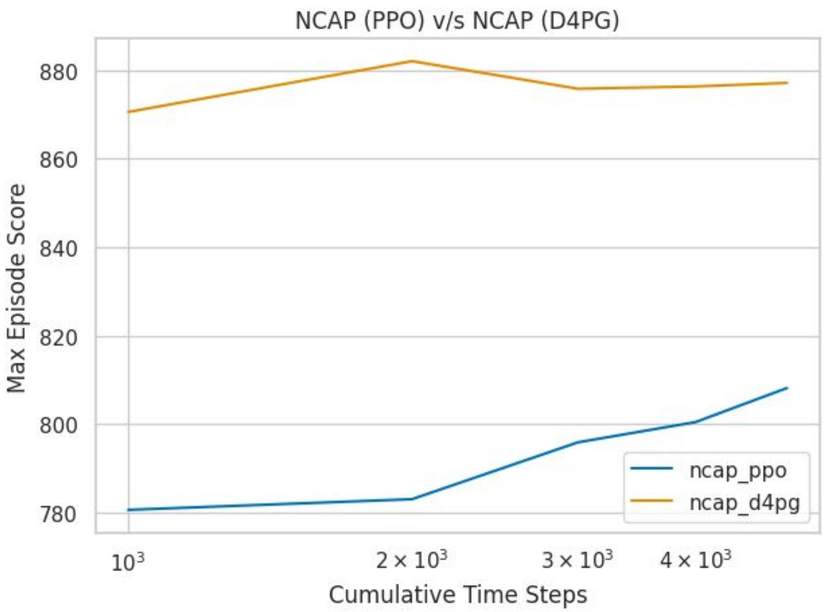
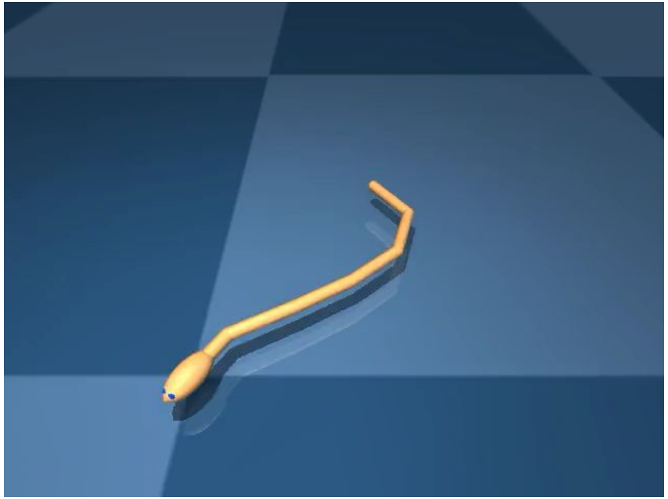
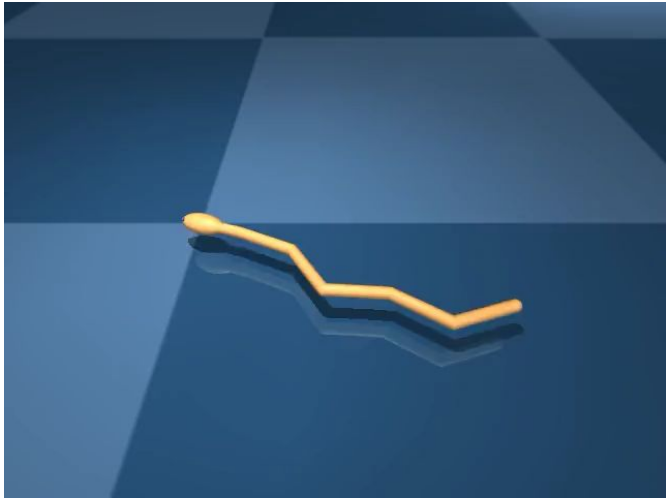




NCAP (PPO)



NCAP (D4PG)

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Conclusion

As a preliminary conclusion, systems with more joints seem to produce more energy efficient closed-loop systems; it is not if this is the result of the interaction with the environment, or the fact the more joint means more redundant actuators, hence more possibilities to allocated energy that may be learn by the MLP without the need to include a control effort constraint in the reward function.

Similarly, the magnitude of the first Fourier moment (i.e. resonant frequency) is higher for the nominal reward function, but this gap seem to reduce as the number of joint is increased.

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